

WEEK 4 REPORT

Exploratory Data Analysis of Road Accidents

Dataset: AccidentsBig_cleaned_week3.csv

Analysis: Univariate and Bivariate Analysis

1. Project Status

Week 4 focuses on Exploratory Data Analysis (EDA) of the cleaned road-accident dataset prepared during Week 3. The analysis is divided into univariate and bivariate analysis, supported by statistical summaries, frequency tables and visualizations.

2. Objectives

- Understand the distribution of important accident-related variables.
- Study accident patterns across severity, road, environmental and temporal variables.
- Examine relationships between pairs of variables using comparative plots and tables.
- Identify useful patterns that can guide later analysis and visualization stages.

3. Dataset Variables Used

Analysis Area	Variables
Accident characteristics	Accident Severity, Number of Vehicles, Number of Casualties
Temporal	Day of Week, Time / Hour
Road characteristics	Road Type, Speed Limit
Environment	Weather Conditions, Light Conditions, Road Surface Conditions
Location / context	Urban or Rural Area, Longitude, Latitude

4. Univariate Analysis

Univariate analysis examines one variable at a time. Frequency distributions, count plots and distribution plots were used to understand the frequency and spread of important variables.

Variable	Analysis Performed
Accident Severity	Frequency distribution
Number of Vehicles	Descriptive statistics and histogram
Number of Casualties	Descriptive statistics and histogram
Day of Week	Frequency distribution
Speed Limit	Frequency distribution
Road Type	Frequency distribution
Weather Conditions	Frequency distribution

Variable	Analysis Performed
Light Conditions	Frequency distribution
Road Surface Conditions	Frequency distribution
Urban / Rural Area	Frequency distribution
Hour of Day	Hourly accident distribution

5. Bivariate Analysis

Bivariate analysis was used to examine relationships between two variables. Cross-tabulations, boxplots, scatter plots, bar charts and line plots were used depending on the variable types.

Variable Pair	Method
Accident Severity × Number of Casualties	Grouped statistics and boxplot
Number of Vehicles × Number of Casualties	Correlation and scatter plot
Weather Conditions × Accident Severity	Cross-tabulation and grouped bar chart
Light Conditions × Accident Severity	Cross-tabulation and grouped bar chart
Road Surface Conditions × Accident Severity	Cross-tabulation and grouped bar chart
Speed Limit × Accident Severity	Cross-tabulation and grouped bar chart
Urban / Rural Area × Accident Severity	Cross-tabulation and grouped bar chart
Day of Week × Number of Casualties	Grouped statistics and mean bar chart
Hour of Day × Accident Severity	Cross-tabulation and line chart
Selected numerical variables	Correlation matrix and heatmap

6. Visualizations Produced

- Count plots for categorical and coded categorical variables.
- Histograms for Number of Vehicles and Number of Casualties.
- Boxplot comparing Accident Severity with Number of Casualties.
- Scatter plot for Number of Vehicles versus Number of Casualties.
- Grouped bar charts for environmental and road variables against Accident Severity.
- Hourly accident distribution and hourly severity line chart.
- Correlation heatmap for selected numerical variables.

7. Interpretation Framework

The analysis is intended to describe patterns rather than establish causal relationships. Categorical variables are represented using the codes stored in the cleaned dataset. Frequency differences show how observations are distributed across categories, while bivariate comparisons show how two variables vary together. Correlation values indicate statistical association between selected numerical variables and should not be interpreted as proof of causation.

8. Week 4 Outcome

Week 4 establishes the project's exploratory analysis layer. The cleaned Week 3 dataset is examined from both single-variable and two-variable perspectives. The resulting tables and visualizations provide a structured basis for subsequent analysis, feature engineering or dashboard development.

9. Conclusion

Univariate analysis provides an overview of accident distributions across severity, casualties, vehicles, road conditions, environmental conditions and time. Bivariate analysis extends this by examining relationships among accident severity, casualties, road characteristics, environmental conditions and temporal factors.

Together, these analyses prepare the dataset for more advanced data storytelling and visualization in subsequent project stages.

Formatting: Black-and-white / grayscale only. No colored elements are used in this report.